Vellore Institute of Technology
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PROJECT REPORT

Design and Implementation of a Smart Parking Management System using Verilog HDL

Submitted by :

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1.INTRODUCTION

• In the urban setting, the increasing number of vehicles has posed a significant challenge to efficient parking space management. Traditional parking lots rely on manual monitoring, which translates to confusion, high waiting times, space mismanagement, and congestion at entry/exit points. Users do not know about available parking spaces in real-time, causing unnecessary movement of vehicles within the parking lot and high fuel consumption.

• Thus, there is a requirement for a digital Smart Parking System that has the ability to automatically track parking spaces, capture vehicle entry and exit, update available space numbers, and deny entry when the parking facility fills up. The system must be simple, reliable, and implementable through the principles of Digital System Design and Verilog HDL. The aim is to automate parking functions, minimize human intervention, and display real-time slot availability data.

1.1 Literature Review

Smart parking systems have been introduced to address the inefficiencies of traditional parking management, which often rely on manual monitoring. Such systems lead to delays, confusion, and poor utilization of parking spaces, especially in urban areas. Researchers have recognized that automating parking operations can help reduce congestion and improve the overall efficiency of vehicle management.

Several studies in intelligent transportation systems highlight the benefits of integrating automation with real-time monitoring. Idris et al. (2009) and Tang et al. (2006) showed that systems capable of tracking slot availability and controlling entry–exit gates automatically provide smoother traffic flow and better user convenience. These findings laid the foundation for smart parking technologies.

Earlier implementations often used camera or IoT-based approaches for slot detection. While effective, such systems can be expensive and complex to maintain. In contrast, digital design using Verilog HDL offers a simpler, low-cost, and hardware-efficient solution that ensures reliable and real-time operation suitable for compact systems.

Verilog-based smart parking designs use counters, registers, and control logic to automate vehicle entry, exit, and slot management. The literature supports that hardware-level automation provides higher speed, lower power consumption, and better scalability.

1.2 AIM

To design and develop a **Smart Parking Management System using Verilog HDL.**

1.3 Overview / Brief of the Project

The **Smart Parking Management System** is designed to **automate parking operations** using **Verilog HDL**. It detects vehicle entry and exit, allocates or frees slots automatically, counts available spaces, and indicates when the parking is full. The design uses digital logic, counters, and control circuits to simulate real-time parking management efficiently

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2. Materials / Components Used

Since it's a digital design project (Verilog-based), the main components include:

- **FPGA Board / Simulator** (e.g., Xilinx Vivado or ModelSim)
- **Verilog HDL Code**
- **Digital Sensors (Entry/Exit Detection)** – simulated inputs
- **LED Indicators** – for gate control and "FULL" signal
- **Registers and Counters (in code)** – for slot allocation logic
- **Clock and Reset Inputs**

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3. SOFTWARE CODE:

```
module smart_parking(
   input clk, rst,
   input entry, exit,
   output reg entry_gate, exit_gate,
   output reg [3:0] free_count,
   output reg full
);

   reg [7:0] slots; // 1 = occupied, 0 = free
   integer i;

   always @(posedge clk or posedge rst) begin
      if (rst) begin
         slots <= 8'b00000000;   // all slots free
         free_count <= 8;
         full <= 0;
         entry_gate <= 0;
         exit_gate <= 0;
      end else begin
         entry_gate <= 0;
         exit_gate <= 0;

         // Car Entry
         if (entry && free_count > 0) begin
            for (i = 0; i < 8; i = i + 1) begin
               if (slots[i] == 0) begin
                  slots[i] <= 1;
                  free_count <= free_count - 1;
                  entry_gate <= 1;
                  i = 8; // exit loop safely
               end
```

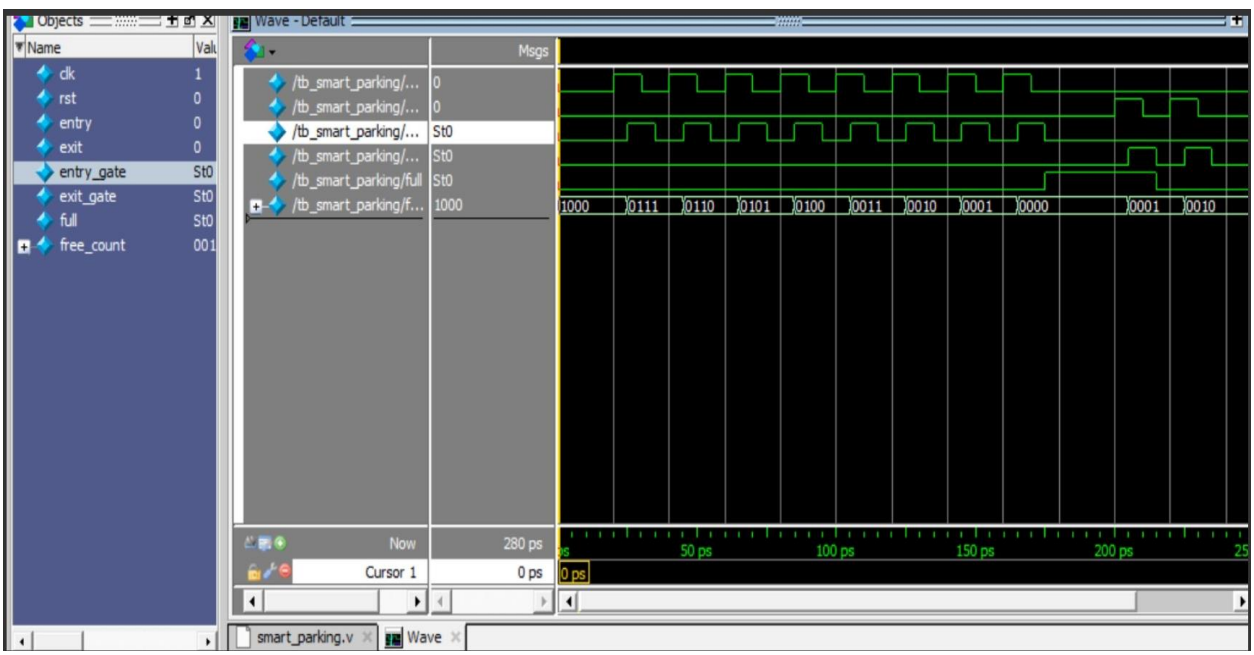
```
            end
        end

        // Car Exit
        if (exit && free_count < 8) begin
            for (i = 7; i >= 0; i = i - 1) begin
                if (slots[i] == 1) begin
                    slots[i] <= 0;
                    free_count <= free_count + 1;
                    exit_gate <= 1;
                    i = -1; // exit loop safely
                end
            end
        end

        // FULL check
        full <= (free_count == 0) ? 1 : 0;
    end
  end
endmodule

module tb_smart_parking;

  reg clk, rst, entry, exit;
  wire entry_gate, exit_gate, full;
  wire [3:0] free_count;

  // Instantiate the Smart Parking module
  smart_parking uut (
    .clk(clk),
    .rst(rst),
    .entry(entry),
    .exit(exit),
```

```
    .entry_gate(entry_gate),
    .exit_gate(exit_gate),
    .free_count(free_count),
    .full(full)
  );

  // Clock generation (10ns period)
  always #5 clk = ~clk;

  initial begin
    clk = 0;
    rst = 1;
    entry = 0;
    exit = 0;

    #10 rst = 0;  // Release reset

    // Simulate car entries
    #10 entry = 1; #10 entry = 0;
    #10 entry = 1; #10 entry = 0;
    #10 entry = 1; #10 entry = 0;
    #10 entry = 1; #10 entry = 0;
    #10 entry = 1; #10 entry = 0;
    #10 entry = 1; #10 entry = 0;
    #10 entry = 1; #10 entry = 0;
    #10 entry = 1; #10 entry = 0; // Now full

    #20; // Wait to observe full signal
// Simulate car exits
    #10 exit = 1; #10 exit = 0;
    #10 exit = 1; #10 exit = 0;

    #50 $stop; // End simulation
```

```
    end
endmodule
```

4. SIMULATION RESULTS

- Initially: free_count = 8, all slots empty.
- Each entry → free_count decreases by 1, entry_gate = 1.
- When all slots filled → full = 1.
- Each exit → free_count increases, exit_gate = 1, slot freed.
- Waveforms confirm correct functionality



5. Block Diagram

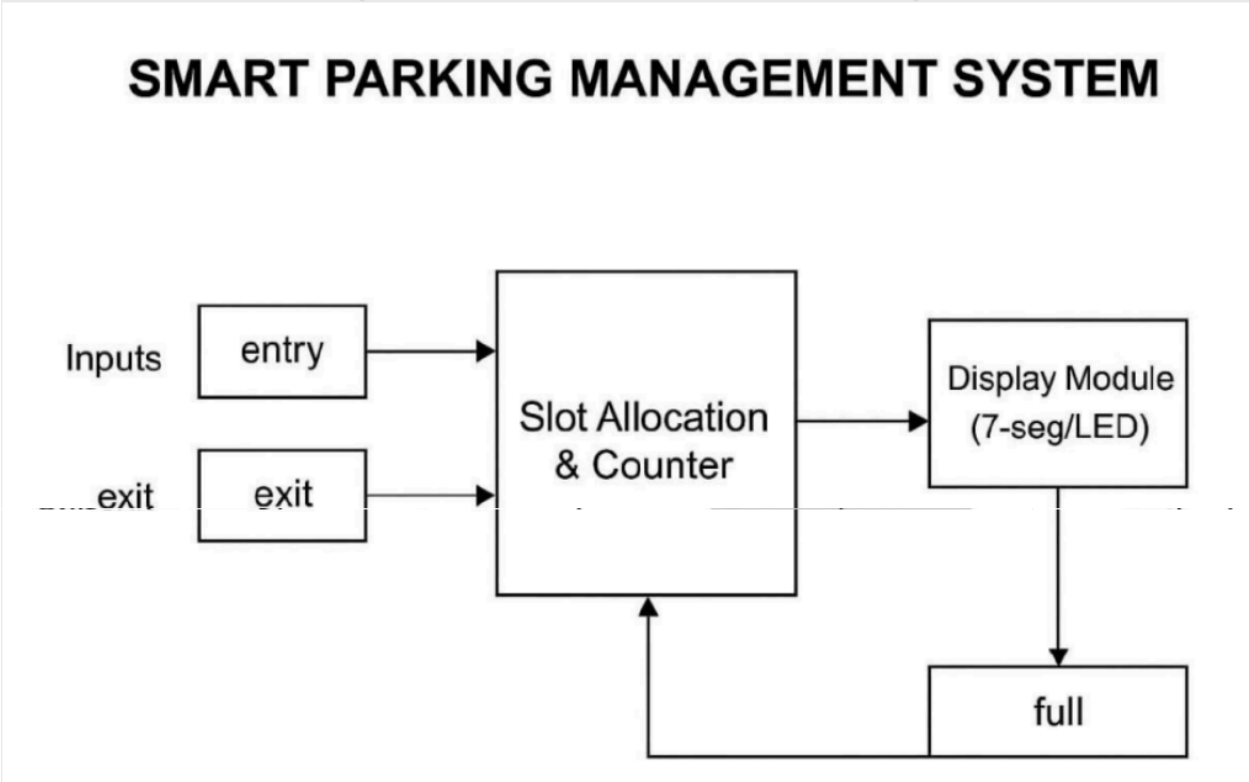
Inputs:
`clk`, `rst`, `entry`, `exit`

Outputs:
`entry_gate`, `exit_gate`, `free_count`, `full`

Processing Unit:

Slot allocation logic that:

- Decrements counter on entry
- Increments counter on exit
- Activates “FULL” signal when all slots are occupied



6. Future Scope

The project can evolve into a full smart-city–grade parking system:

1. **IoT Integration:** View slot availability via mobile apps.
2. **Automated Billing:** Time-based fare deduction and ticketing.
3. **ANPR System:** Vehicle number plate recognition for security.
4. **Sensor-based Slot Detection:** Using IR/Ultrasonic sensors for accuracy.
5. **Multi-Level Parking Support:** Expand logic to handle multiple floors.
6. **Guidance System:** LEDs or voice alerts to guide drivers to free slots.
7. **Priority/Emergency Parking:** Reserved zones for specific vehicles.

7. CONCLUSION

The Smart Parking Management System was successfully modeled and simulated with Verilog HDL. The system effectively tracks availability of parking slots, automates the car entry-exit process, and guards against overfilling by tripping a full indicator. Simulation waveforms confirm that counter counts down on each car entry and up on exit, with full tripping at zero availability. This project illustrates how digital logic, counters, and control circuits are efficiently employed to address actual-world automation issues. The design is intuitive, expandable, and can be applied on FPGA hardware or developed into an IoT-based system for contemporary Smart City implementation. The outcomes confirm that the system is robust, affordable, and able to enhance parking efficiency with minimal manual intervention

8. REFERENCES

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